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Week 14 - Full Results Analysis

Interpretation of the final reconstructed cohort and robustness checks

1. Cohort

The analysis starts with 2,657 sprint rows. Thirty FUTURE rows, 14 ACTIVE rows, and three CLOSED rows with impossible timestamps are removed before label construction. One first eligible sprint per project is removed because there is no prior history. The final model frame has 2,602 sprints.

Step	Count
Raw sprint rows	2,657
Eligible closed sprints	2,610
Cold starts removed	8
Modeled sprints	2,602
Reconstructed commitments	20,603
Modeled zero-scope sprints	911
Modeled nonempty sprints	1,691

2. Main results

Late closure remains a moderate prediction problem. The main random-forest AUC is .705. Conditional spillover is easier to rank, but the transparent scope-only logistic model reaches .885 AUC, compared with .852 for random forest. The result favors parsimony: initial scope explains most of the measurable spillover signal.

Outcome / model	Test n	Positives	F1	ROC-AUC
Late closure - Random Forest	786	246	.562	.705
Late closure - XGBoost	786	246	.535	.695
Nonempty spillover - Always positive	511	482	.971	.500
Nonempty spillover - Scope-only logistic	511	482	.696	.885
Nonempty spillover - Random Forest	511	482	.875	.852

3. Uncertainty

Project-cluster intervals are wide enough that small model differences should not be overinterpreted. Late-closure random forest has a 95% AUC interval of .604-.783. Conditional spillover has .820-.922 for scope-only logistic regression and .797-.914 for random forest.

4. Temporal and cross-project stability

Forward cut points give late-closure RF AUCs from .669 to .711 and conditional-spillover RF AUCs from .846 to .856. Held-out-project performance varies much more, especially for late closure. The evidence supports local validation rather than a single global performance claim.

5. Provenance sensitivity

Static fallback is an identification assumption whenever no change history is recorded. Requiring observed Sprint history for every committed item leaves 356 test sprints and gives .875 AUC for the scope-only model and .850 for random forest. Requiring both Sprint and Story Points history leaves only 110 test sprints; AUC falls to .772 and .652 respectively.

Subset	Test n	Scope LR AUC	RF AUC
Main nonempty cohort	511	.885	.852
Observed Sprint history	356	.875	.850
Observed Sprint + Story Points history	110	.772	.652

6. Feature evidence

Held-out permutation importance shows committed issue count is the dominant conditional-spillover feature, with an average AUC drop of .220 when shuffled. For late closure, project identity and sprint length are the largest contributors.

7. Interpretation

The strongest conclusion is not that a sophisticated model predicts Scrum success. It is that reconstructed sprint-start scope carries useful information about unfinished work, and a simple scope model captures most of it. Late closure is weaker and more project-specific. The current evidence is retrospective and should not be presented as proof that exposing scores to Scrum teams improves delivery.